

Integrating hyperbolic trig functions

Basic skills

$$\text{Q1. } \frac{d}{dx} (2 \sinh(4x)) = 2 \times 4 \cosh 4x \\ = 8 \cosh x$$

$$\frac{d}{dx} (\sinh x) = \cosh x$$

$$\text{Q2. } y = \tanh x \Rightarrow \frac{dy}{dx} = \operatorname{sech}^2 x$$

$$\text{so at } x=0 \quad \frac{dy}{dx} = \operatorname{sech}^2 0 = \frac{1}{\cosh^2 0} = \frac{1}{1^2} = 1$$

$$\text{Q3. } \int e^{kx} dx = \frac{1}{k} e^{kx}$$

$$\text{Q4. } \frac{d}{dx} (\cosh(\underbrace{4-x^2}_u)) = \frac{d}{du} (\cosh u) \frac{d}{dx} (\overbrace{4-x^2}^u) \\ = -2x \sinh u = -2x \sinh(4-x^2)$$

$$\text{Q5. } \int \sinh 2x dx = \frac{1}{2} \cosh 2x$$

$$\int \sinh x dx = \cosh x$$

Q6. $\int \cosh x \, dx = \sinh x$ or $\int \frac{e^x + e^{-x}}{2} \, dx$
 $= \frac{e^x - e^{-x}}{2}$
 $= \sinh x$

as $\frac{d}{dx} (\sinh x) = \cosh x$

Q7. $\int \sinh x \, dx = \int \frac{e^x - e^{-x}}{2} \, dx$
 $= \frac{e^x + e^{-x}}{2} = \cosh x$

Q8. $\int_1^2 \operatorname{sech}^2 x \, dx = [\tanh x]_1^2$
 $= \tanh 2 - \tanh 1$

Competency skills

Q9. a) $\int 2 \cosh 5x \, dx = \frac{2}{5} \sinh 5x$

b) $\int x \sinh 2x \, dx = \frac{1}{2} x \cosh 2x - \int \frac{1}{2} \cosh 2x \, dx$
Parts
 $u = x \quad v = \frac{1}{2} \cosh 2x$
 $u' = 1 \quad v' = \sinh 2x$
 $= \frac{1}{2} x \cosh 2x - \frac{1}{4} \sinh 2x$

c) $\int x \operatorname{sech}^2(x^2 + 1) \, dx$ Sub $u = x^2 + 1$
 $u' = 2x$

Then $dx = \frac{1}{2x} du$ so

$$\int x \operatorname{sech}^2 u \times \frac{1}{2x} du = \frac{1}{2} \int \operatorname{sech}^2 u du$$

$$= \frac{1}{2} \tanh u = \frac{1}{2} \tanh(x^2 + 1)$$

Q10. $\int \operatorname{arccosh} x dx$

Let $u = \operatorname{arccosh} x$

$v = x$

$v' = 1$

$u' = \frac{1}{\sqrt{x^2 - 1}}$

USE formula or

$x = \cosh u$

$\frac{dx}{du} = \sinh u = \sqrt{\cosh^2 u - 1} = \sqrt{x^2 - 1}$ so $\frac{du}{dx} = \frac{1}{\sqrt{x^2 - 1}}$

so $\int \operatorname{arccosh} x dx = x \operatorname{arccosh} x - \int \frac{x}{\sqrt{x^2 - 1}} dx$

$= x \operatorname{arccosh} x - \int \frac{1}{2\sqrt{u}} du$

$u = x^2 - 1$
 $du = 2x dx$

$= x \operatorname{arccosh} x - \frac{1}{2} \times \frac{u^{\frac{1}{2}}}{\frac{1}{2}}$

$= x \operatorname{arccosh} x - u^{\frac{1}{2}} = x \operatorname{arccosh} x - \sqrt{x^2 - 1}$

Q11. $\int_0^2 \operatorname{arcsinh} x \, dx$

By parts $u = \operatorname{arcsinh} x$ $v = x$

$u' = \frac{1}{\sqrt{x^2+1}}$ $v' = 1$

formula or
method $x = \cosh u$

so $\int \operatorname{arcsinh} x \, dx = x \operatorname{arcsinh} x - \int \frac{x}{\sqrt{x^2+1}} \, dx$

$= x \operatorname{arcsinh} x - \int \frac{1}{2\sqrt{u}} \, du$ $\swarrow u = x^2+1$

$= x \operatorname{arcsinh} x - \frac{1}{2} \frac{u^{\frac{1}{2}}}{\frac{1}{2}}$

$= x \operatorname{arcsinh} x - \sqrt{x^2+1}$

so $\int_0^2 \operatorname{arcsinh} x \, dx = 2 \operatorname{arcsinh} 2 - \sqrt{2^2+1}$
 $- (0 - \sqrt{0^2+1})$

$= 2 \operatorname{arcsinh} 2 - \sqrt{5} + 1$

Q12. $\int \operatorname{cosech} x \, dx = \int \frac{2}{e^x - e^{-x}} \, dx$

$= \int \frac{2e^x}{e^{2x}-1} \, dx = \int \frac{2u}{u^2-1} \, dx$ $u = e^x$
 $u' = e^x = u$

so $dx = \frac{1}{u} du$

so $= \int \frac{2u}{u^2-1} \frac{1}{u} du = \int \frac{2}{u^2-1} du$

$= \int \frac{1}{u-1} - \frac{1}{u+1} du$

partial fractions

$$\frac{2}{u^2-1} = \frac{A}{u-1} + \frac{B}{u+1}$$

$= \ln(u-1) - \ln(u+1)$

so $A+B=0$

$A-B=2$

$A=1, B=-1$

$= \ln\left(\frac{u-1}{u+1}\right) = \ln\left(\frac{e^x-1}{e^x+1}\right)$

$= \ln\left(\frac{e^{\frac{x}{2}} - e^{-\frac{x}{2}}}{e^{\frac{x}{2}} + e^{-\frac{x}{2}}}\right) = \ln\left(\tanh\left(\frac{x}{2}\right)\right)$

Logarithm input needs to be positive so

$\ln\left|\tanh\left(\frac{x}{2}\right)\right|$

Q13. $\int_{-2}^2 \cosh^2 x dx$



Now use identity

method is like that of $\cos^2 x$

$\cosh 2x = \cosh^2 x + \sinh^2 x$

$= \cosh^2 x + (\cosh^2 x - 1) = 2\cosh^2 x - 1$

$$\text{SO } \cosh^2 x = \frac{1}{2} + \frac{1}{2} \cosh 2x$$

$$\int_{-2}^2 \left(\frac{1}{2} + \frac{1}{2} \cosh 2x \right) dx = \left[\frac{1}{2}x + \frac{1}{4} \sinh 2x \right]_{-2}^2$$

$$= \left(1 + \frac{1}{4} \sinh 4 \right) - \left(-1 + \frac{1}{4} \sinh(-4) \right)$$

$$= 2 + \frac{1}{4} (\sinh 4 - \sinh(-4))$$

$$= 2 + \frac{1}{2} \sinh 4$$

✗ $\sinh(-x) = -\sinh x$

Advanced skills

Q14. a) $\int_a^b \coth x \, dx = \int_a^b \frac{\cosh x}{\sinh x} \, dx$

$$= \left[\ln |\sinh x| \right]_a^b$$

$\int \frac{f'(x)}{f(x)} \, dx = \ln |f(x)|$

$$= \ln |\sinh b| - \ln |\sinh a| = \ln \left(\frac{|\sinh b|}{|\sinh a|} \right)$$

b) As $\coth x$ is undefined at $x=0$
split up

$$\int_{-2}^2 \coth x \, dx = \int_0^2 \coth x \, dx + \int_{-2}^0 \coth x \, dx$$

but if $a=0$ our expression doesn't work as " $\ln\left(\frac{\sinh 2}{\sinh 0}\right) = \ln\left(\frac{\sinh 2}{0}\right)$ "

so the improper integral does not exist

$$\begin{aligned} c) \int_1^{\infty} \coth x \, dx &= \lim_{k \rightarrow \infty} \int_1^k \coth x \, dx \\ &= \lim_{k \rightarrow \infty} \ln\left(\frac{|\sinh k|}{|\sinh 1|}\right) \quad \text{as } k \rightarrow \infty \quad \sinh k \rightarrow \infty \\ &\quad \text{diverges} \\ &= \infty \end{aligned}$$

Area does not exist.

Q15. $\int x^3 \cosh x^2 \, dx$ let $u = x^2$

$$\frac{du}{dx} = 2x$$

$$= \int x^3 \cosh u \cdot \frac{1}{2x} \, du$$

$$= \int \frac{1}{2} u \cosh u \, du$$

$$= \frac{1}{2} (u \sinh u - \int \sinh u \, du)$$

now parts

$$m = u$$

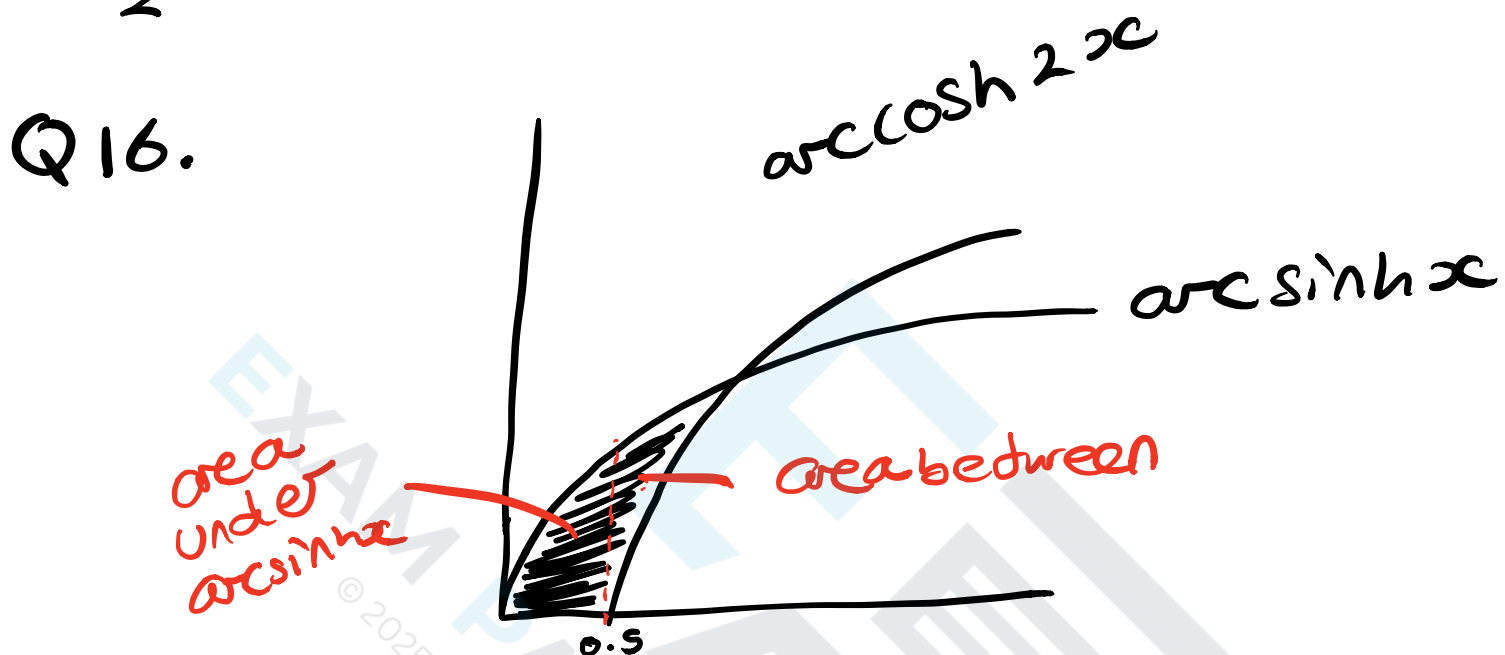
$$n = \sinh u$$

$$m' = 1$$

$$n' = \cosh u$$

$$= \frac{1}{2} u \sinh u - \frac{1}{2} \cosh u$$

$$= \frac{1}{2} x^2 \sinh x^2 - \frac{1}{2} \cosh x^2$$



For point of equality

$$\text{arc cosh } 2x = \text{arc sinh } x$$

$$\Rightarrow \ln(2x + \sqrt{4x^2 - 1}) = \ln(x + \sqrt{x^2 + 1})$$

$$\Rightarrow 2x \pm \sqrt{4x^2 - 1} = x + \sqrt{x^2 + 1}$$

$$\Rightarrow x \pm \sqrt{4x^2 - 1} = \sqrt{x^2 + 1}$$

$$\Rightarrow x^2 \pm 2x\sqrt{4x^2 - 1} + 4x^2 - 1 = x^2 + 1$$

$$\Rightarrow 4x^2 \pm 2x\sqrt{4x^2 - 1} - 2 = 0$$

$$\Rightarrow 4x^2 - 2 = \pm 2x\sqrt{4x^2 - 1}$$

$$\text{SO } (4x^2 - 2)^2 = (2x\sqrt{4x^2 - 1})^2$$

$$\Rightarrow 16x^4 - 16x^2 + 4 = 4x^2(4x^2 - 1)$$

$$\Rightarrow 16x^4 - 16x^2 + 4 = 16x^2 - 4x^2$$

$$\Rightarrow 12x^2 = 4 \Rightarrow x^2 = \frac{1}{3}$$

$$\Rightarrow x = \frac{1}{\sqrt{3}}$$

SO between $\frac{1}{\sqrt{3}}$ and $\frac{1}{2}$ is $\int_{\frac{1}{2}}^{\frac{1}{\sqrt{3}}} (\operatorname{arcsinh} x - \operatorname{arccosh} 2x) dx$

USE parts $\begin{matrix} 1 \times \operatorname{arc} \\ \text{Int} \\ \text{this} \end{matrix}$ $\begin{matrix} \operatorname{arc} \\ \text{Diff} \\ \text{this} \end{matrix}$

$$u = \operatorname{arcsinh} x \quad v = x$$

$$u' = \frac{1}{\sqrt{x^2 + 1}} \quad v' = 1$$

formula

$$u = \operatorname{arccosh} 2x \quad v = x$$

$$u' = \frac{1}{\sqrt{x^2 - \frac{1}{4}}} \quad v' = 1$$

formula

$$= \left[x \operatorname{arcsinh} x \right]_{\frac{1}{2}}^{\frac{1}{\sqrt{3}}} - \int_{\frac{1}{2}}^{\frac{1}{\sqrt{3}}} \frac{x}{\sqrt{x^2 - 1}} dx$$

Sub $u = x^2 - 1$

$$- \left(\left[x \operatorname{arccosh} 2x \right]_{\frac{1}{2}}^{\frac{1}{\sqrt{3}}} - \int_{\frac{1}{2}}^{\frac{1}{\sqrt{3}}} \frac{2x}{\sqrt{4x^2 - 1}} dx \right)$$

Sub $u = 4x^2 - 1$

$$= \left[x \operatorname{arcsinh} x - \sqrt{x^2 + 1} \right]_{\frac{1}{2}}^{\frac{1}{\sqrt{3}}}$$

$$- \left[x \operatorname{arccosh} 2x - \frac{1}{2} \sqrt{4x^2 - 1} \right]_{\frac{1}{2}}^{\frac{1}{\sqrt{3}}}$$

Now other area is

$$\int_0^{\frac{1}{2}} \operatorname{arcsinh} x \, dx = \left[x \operatorname{arcsinh} x - \sqrt{x^2 + 1} \right]_{\frac{1}{2}}^{\frac{1}{\sqrt{3}}}$$

so

$$\begin{aligned} \text{Total} &= \left[x \operatorname{arcsinh} x - \sqrt{x^2 + 1} \right]_0^{\frac{1}{\sqrt{3}}} \\ &\quad - \left[x \operatorname{arccosh} 2x - \frac{1}{2} \sqrt{4x^2 - 1} \right]_{\frac{1}{2}}^{\frac{1}{\sqrt{3}}} \end{aligned}$$

$$\begin{aligned} &= \frac{1}{\sqrt{3}} \operatorname{arcsinh} \left(\frac{1}{\sqrt{3}} \right) - \sqrt{\frac{4}{3}} - (0 - 1) \\ &\quad - \left(\frac{1}{\sqrt{3}} \operatorname{arccosh} \left(\frac{2}{\sqrt{3}} \right) - \frac{1}{2} \sqrt{\frac{1}{3}} - \left(\frac{1}{2} \operatorname{arccosh} 1 - 0 \right) \right) \end{aligned}$$

$$= \frac{1}{\sqrt{3}} \left(\operatorname{arcsinh} \left(\frac{1}{\sqrt{3}} \right) - \operatorname{arccos} \left(\frac{2}{\sqrt{3}} \right) \right) + 1 - \frac{3}{2\sqrt{3}}$$

$$= \frac{1}{\sqrt{3}} \left(\ln \left(\frac{1}{\sqrt{3}} + \sqrt{\frac{4}{3}} \right) - \ln \left(\frac{2}{\sqrt{3}} + \sqrt{\frac{1}{3}} \right) \right) + 1 - \frac{3}{2\sqrt{3}}$$

$$= \frac{1}{\sqrt{3}} \ln \left(\frac{\frac{3}{\sqrt{3}}}{\frac{3}{\sqrt{3}}} \right) + 1 - \frac{3}{2\sqrt{3}} = 0 + 1 - \frac{3}{2\sqrt{3}} = \frac{2 - \sqrt{3}}{2}$$